

Soroush Vahidi
New Jersey Institute of Technology
Newark, NJ, USA
sv96@njit.edu
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Editor-in-Chief
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Dear Editor,

Thank you for the opportunity to revise my manuscript, now entitled “*Training-Free Ranking from Pairwise Comparisons via Acyclic Graph Construction*,” and for the careful reviews for *The Journal of Supercomputing*. I am submitting the revised manuscript for reconsideration. The manuscript is original, has not been published previously, and is not under consideration elsewhere. I take full responsibility for the scientific content of this revision.

The manuscript has undergone a substantial major revision in response to all four reviewers. The accompanying point-by-point response addresses every comment. The principal changes are:

1. The title and positioning were revised to remove an unqualified scalability claim and to distinguish prior MWFAS components from the contributions of this work.
2. The submitted fixed-topological reinsertion proxy was replaced in the canonical pipeline by exact reachability-aware reinsertion.
3. Principal Tables 4–6 now evaluate the canonical reachability-aware method, OURS-REACH, rather than the submitted topology-proxy variant.
4. The inconsistent baseline tables identified by Reviewer 3 were rebuilt from a single canonical data source.
5. Direct runtime comparisons against training-free classical baselines were added. The manuscript states that the method is slower than most lightweight classical methods, while retaining a substantial end-to-end runtime advantage over the evaluated trained GNN methods.
6. Structural ablations, sensitivity analyses, timeout/coverage analysis, family-aware analysis, and statistical tests were added.
7. Theory was strengthened with ranking–MWFAS optimum-value equivalence, exact reachability correctness, Demetrescu–Finocchi guarantee-boundary clarification, a fallback impossibility result, and a secondary weighted min-cut exchange with guarantees.
8. The manuscript was substantially rewritten to reduce repetition and improve readability.

I am grateful to the editor and reviewers; their comments led to substantial improvements. Detailed responses appear in the accompanying response-to-reviewers document.

Sincerely,

Soroush Vahidi
New Jersey Institute of Technology
sv96@njit.edu